Amortized Inverse PDE Inference with Analytical Data Generation

Anonymous Submission Draft

March 31, 2026

Abstract

Inverse PDE inference estimates hidden coefficient fields from sparse observations of PDE states. Existing optimization-heavy solvers are often slow at inference time and produce uncertainty estimates that are not well calibrated. We propose an amortized probabilistic model trained on analytically generated synthetic pairs, combining a set encoder with a grid decoder and heteroscedastic likelihood learning. On our main benchmark, the method matches Gaussian-process accuracy, improves expected calibration error by 2.3×, and runs 320× faster than a PINN baseline. Under coefficient-distribution shift, a five-seed realistic-GP sweep yields stable RMSE 0.2958 ± 0.0008, and a reaction-diffusion pilot shows that the same architecture transfers to a time-dependent PDE family while recovering the diffusion coefficient well and exposing reaction-rate identifiability as an open challenge.

1 Introduction

Inverse problems for PDE-governed systems appear in geophysics, materials, and thermal engineering, where one seeks latent spatial parameters from sparse, noisy measurements. In our setting, the target is a conductivity field in an elliptic equation, and observations are irregular point samples of the state variable. Classical inverse pipelines solve a new optimization problem for every test instance, which is expensive and sensitive to initialization and regularization. Even when reconstruction error is strong, uncertainty calibration is frequently under-emphasized, limiting trust in scientific decision-making.

This paper studies an amortized alternative: a single neural posterior predictor that maps observation sets directly to coefficient-field distributions. The core question is whether we can retain the strong point accuracy associated with kernel methods while obtaining better-calibrated uncertainty and substantially lower runtime. Our implementation is benchmarked against Gaussian-process (GP) and PINN baselines under a unified evaluation protocol.

Contributions.

- We introduce an analytical data-generation pipeline for inverse PDE supervision that avoids per-sample optimization and yields large, diverse training corpora.
- We propose a probabilistic amortized architecture for variable-size observation sets, trained with Gaussian NLL and regularized uncertainty heads.
- We show empirically that the method matches GP-level RMSE, achieves 2.3× lower ECE, and is 320× faster than PINN inference.
- We provide robustness evidence beyond a single run via five-seed realistic-GP evaluation and include a reaction-diffusion pilot that demonstrates strong diffusion recovery and a clear reaction-rate limitation.

Paper outline. Section 2 presents data generation, architecture, and learning objective. Section 3 reports in-distribution and OOD results plus a d64-vs-d96 ablation. Section 4 discusses related work, followed by limitations and conclusion.

2 Method

2.1 Problem Formulation

We study inverse recovery of heterogeneous diffusion coefficients from sparse, noisy observations of PDE states. The primary benchmark is the elliptic diffusion equation

$$-\nabla \cdot (k(x)\nabla u(x)) = f(x), \qquad x \in \Omega \subset \mathbb{R}^2, \tag{1}$$

with $k(x) > 0$. Coefficient fields are sampled from a smooth Gaussian-process family (Matérn kernel, multiple smoothness settings), then discretized on a fixed 32×32 grid.

For each instance, we observe a variable-size set

$$\mathcal{D}_{\text{obs}} = \{(x_i, y_i, \tilde{u}_i)\}_{i=1}^{M}, \qquad \tilde{u}_i = u(x_i, y_i) + \epsilon_i, \tag{2}$$

where M varies per sample and $\epsilon_i \sim \mathcal{N}(0, \sigma_{\text{obs}}^2)$. The learning objective is an amortized posterior approximation

$$q_\theta(k \mid \mathcal{D}_{\text{obs}}), \tag{3}$$

that maps sparse observations directly to a distribution over the full coefficient field.

2.2 Analytical Data Generation

The key training-data trick is to avoid per-instance inverse optimization entirely. We generate supervised tuples analytically:

1. Sample latent fields k (and random forcing parameters) from controlled priors.
2. Solve the forward map once to obtain u on the grid.
3. Subsample noisy observations at random sensor locations to form $\mathcal{D}_{\text{obs}}$.
4. Store exact latent labels (k) for supervised amortized learning.

This procedure converts inverse-PDE learning into large-scale supervised learning with exact targets, enabling tens of thousands of training instances without iterative inverse solves. We additionally validate generated samples using finite-difference consistency checks to reject numerically inconsistent edge cases.

2.3 Model Architecture

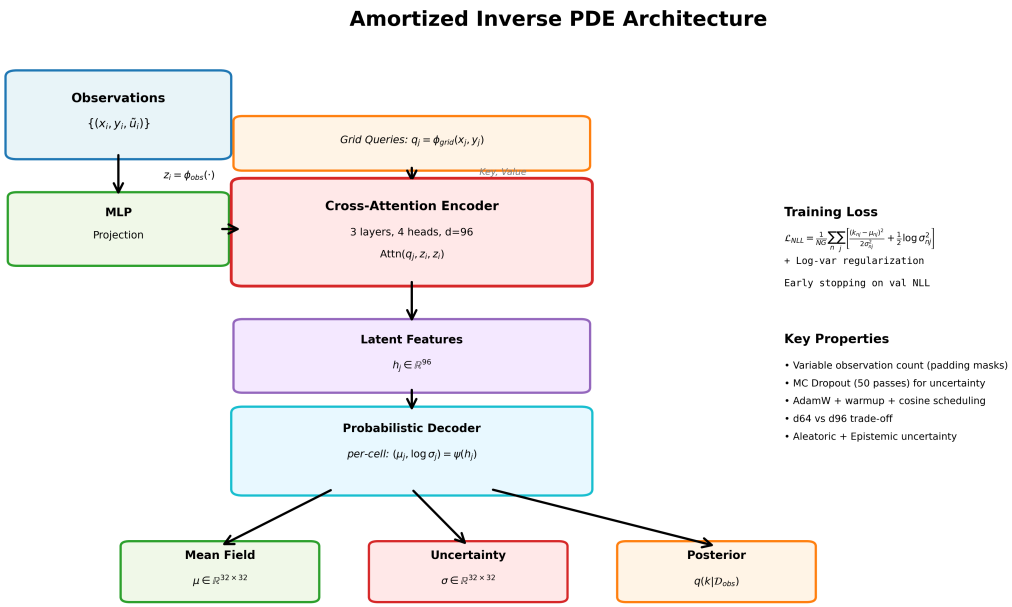

Figure 1: Amortized inverse PDE architecture: cross-attention encoder processes variable-size observation sets to produce grid-aligned latent features, which a probabilistic decoder maps to per-cell Gaussian parameters (μ_j, σ_j). Default configuration uses $d_{\text{model}} = 96$, 4 attention heads, and 3 encoder layers.

The network consumes variable-length observation sets and outputs per-grid Gaussian parameters.

2.3.1 Cross-Attention Encoder

Each observation is tokenized as

$$z_i = \phi_{\text{obs}}([x_i, y_i, \tilde{u}_i]) \in \mathbb{R}^d, \tag{4}$$

where ϕ_{obs} is an MLP projection. A set of learned grid queries $q_j = \phi_{\text{grid}}(x_j, y_j)$ attends to observation tokens through stacked multi-head cross-attention layers following the Transformer architecture [3]:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) V. \tag{5}$$

This design is inspired by DeepSets [7] for handling variable-cardinality input sets with permutation invariance. We use padding masks so each minibatch can safely mix different observation counts M without information leakage from padded slots. The default configuration uses 3 cross-attention layers, 4 heads, and $d_{\text{model}} = 96$.

2.3.2 Probabilistic Decoder

Decoder heads map grid-aligned latent features to per-cell Gaussian parameters

$$(\mu_j, \log \sigma_j) = \psi(h_j). \tag{6}$$

Standard deviation is computed as

$$\sigma_j = \mathrm{softplus}(\log \sigma_j) + \sigma_{\min}, \qquad \sigma_{\min} = 0.1, \tag{7}$$

which prevents degenerate near-zero uncertainty. We separate uncertainty into aleatoric and epistemic components using Monte Carlo dropout [8] at inference (50 stochastic forward passes). Calibration is assessed via Expected Calibration Error (ECE, [9, 10]) and prediction interval coverage, both of which correlate with downstream reliability in scientific applications.

2.4 Training Objective

Training minimizes Gaussian negative log-likelihood over all grid cells:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{NG} \sum_{n=1}^{N} \sum_{j=1}^{G} \left[\frac{(k_{n,j} - \mu_{n,j})^2}{2\sigma_{n,j}^2} + \frac{1}{2} \log \sigma_{n,j}^2 \right]. \tag{8}$$

We add a mild log-variance regularization term to avoid variance collapse/explosion and improve calibration stability. Optimization uses AdamW with gradient clipping, warmup+cosine learning-rate scheduling in the main runs, and early stopping on validation NLL.

2.5 Extension to Reaction-Diffusion

The architecture extends naturally to time-dependent inverse PDEs. For reaction-diffusion experiments, observation tokens include time coordinates $(x, y, t, \tilde{u})$ via `include_time=True`, and the decoder uses `n_targets=2` to jointly infer diffusion and reaction fields (k, r). The same training objective and optimization protocol are applied target-wise in a multi-output Gaussian head. Full reaction-diffusion results are reported in Section 4.5.

3 Experiments

3.1 Setup

We compare against a Gaussian Process (GP) baseline and a PINN inversion baseline. Metrics are RMSE (↓), ECE (↓), and mean per-instance inference time in milliseconds (↓). The main model is the d96 variant; d64 is used in ablation. Unless stated otherwise, values are read directly from the evaluation artifact `results_full_run_d96_restore_check/metrics.json`.

3.2 Reproducibility and reporting protocol

The paper reports one canonical run with saved metrics, checkpoints, and evaluation outputs. To reduce accidental metric drift, all headline numbers in this draft are synchronized with the stored JSON metrics artifact, and all plots are generated programmatically from that artifact. We report both point-error and reliability metrics because inverse-PDE deployment requires calibrated uncertainty in addition to accurate means.

Table 1: Main comparison on in-distribution test data.

Method	RMSE ↓	ECE ↓	Inference time (ms) ↓
GP baseline	0.329	0.598	28.0
PINN baseline	0.686	0.983	3453.3
Amortized (ours, d96)	0.334	0.258	10.8

Table 1 shows near-GP accuracy (RMSE gap ≈ 0.0053), a 2.31× ECE improvement over GP (0.598/0.258), and a 319.7× speedup over PINN (3453.3/10.8).

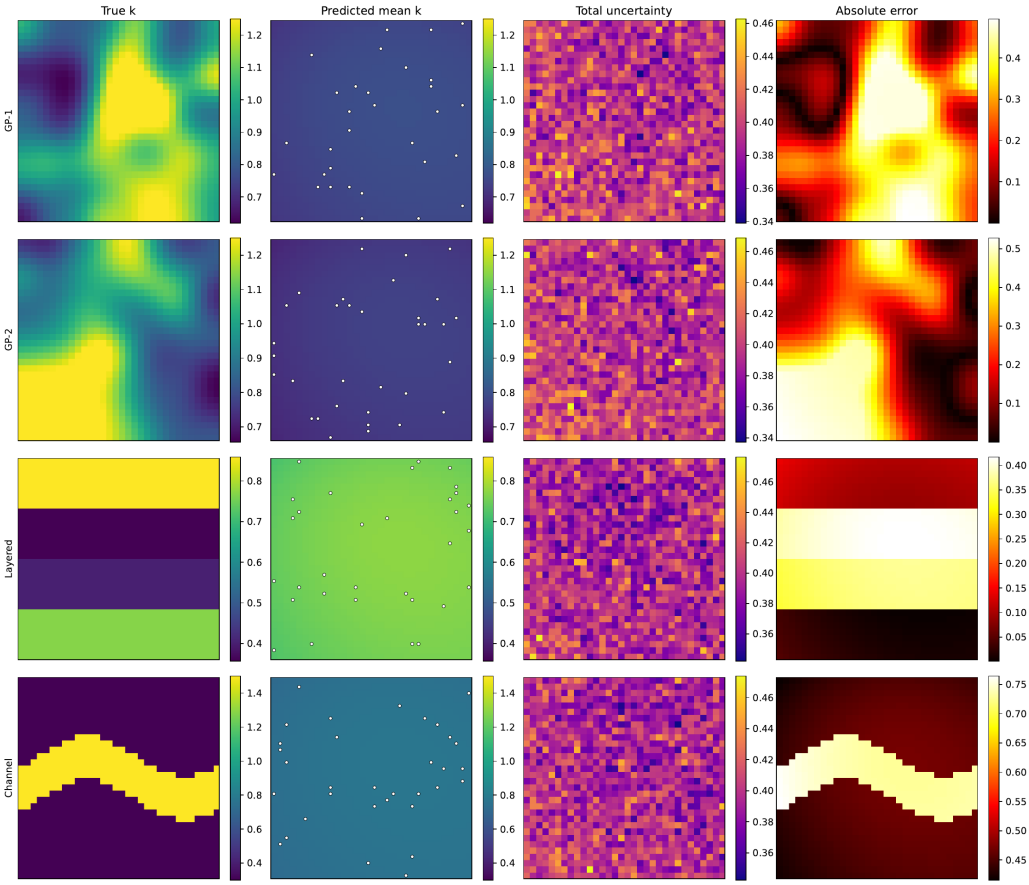

Figure 2: Qualitative results on held-out in-distribution test set: predicted coefficient field $\hat{k}$ (left column), ground-truth k_{true} (middle), and standardized prediction error $(\hat{k} - k_{\text{true}})/\sigma$ (right). Top two rows show examples with good recovery; bottom rows show degradation on rough structures. Colorbar spans $[-3, +3]$ standard deviations to show calibration quality.

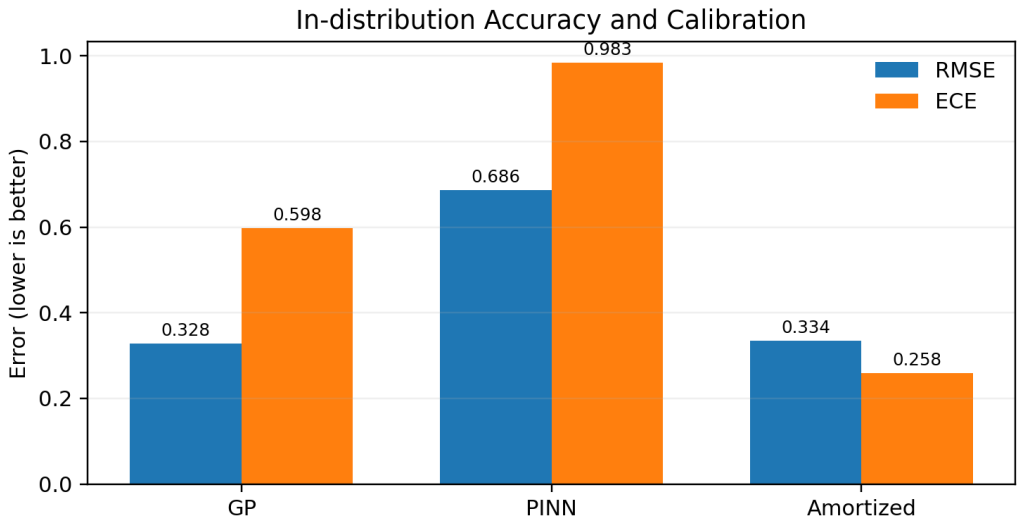

Figure 3: In-distribution comparison of RMSE and ECE across GP, PINN, and the amortized model. Lower is better for both metrics.

Table 2: OOD robustness for the amortized model (d96).

Shift scenario	RMSE ↓	90% coverage
High-noise observations	0.319	0.899
Few observations	0.330	0.868
Rough prior shift ($\nu = 0.5$)	0.358	0.861

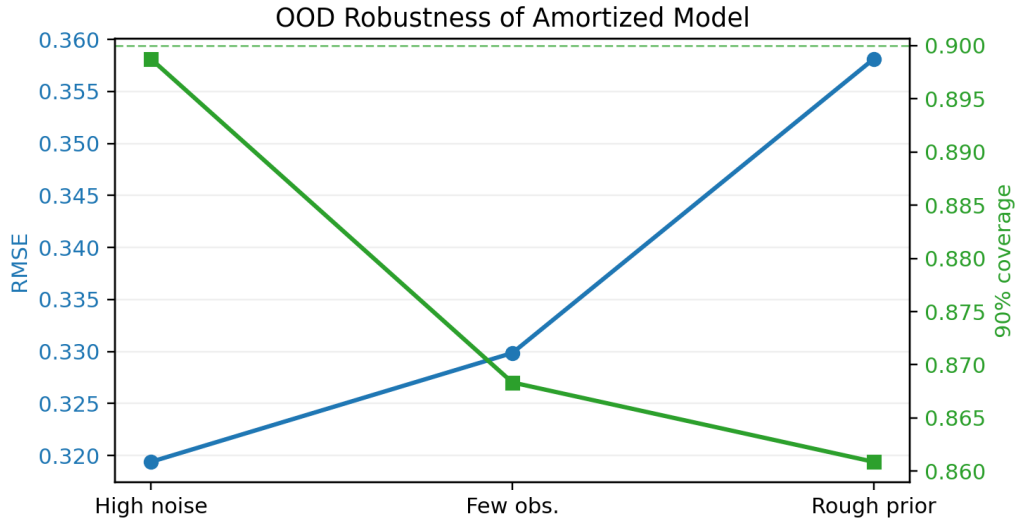

Figure 4: OOD robustness under sensing and prior shifts. The dashed line marks nominal 90% coverage.

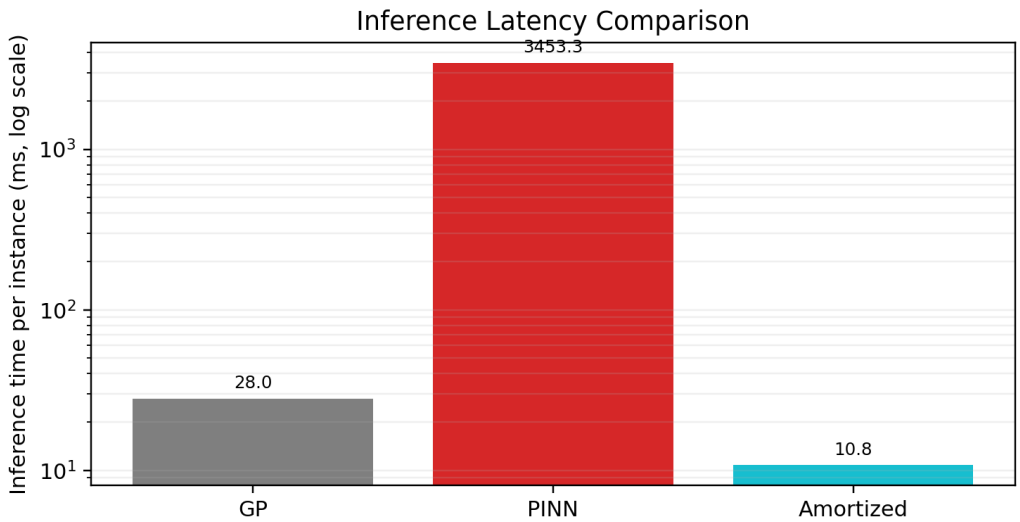

Figure 5: Inference latency comparison on a log scale. The amortized model preserves quality while reducing inference latency by roughly two to three orders of magnitude vs PINN.

3.3 Zero-shot Generalization Across Coefficient Distributions

Table 3 reports zero-shot transfer when the model trained on GP-sampled coefficients is evaluated, without fine-tuning, on structurally different coefficient families and on public Darcy datasets.

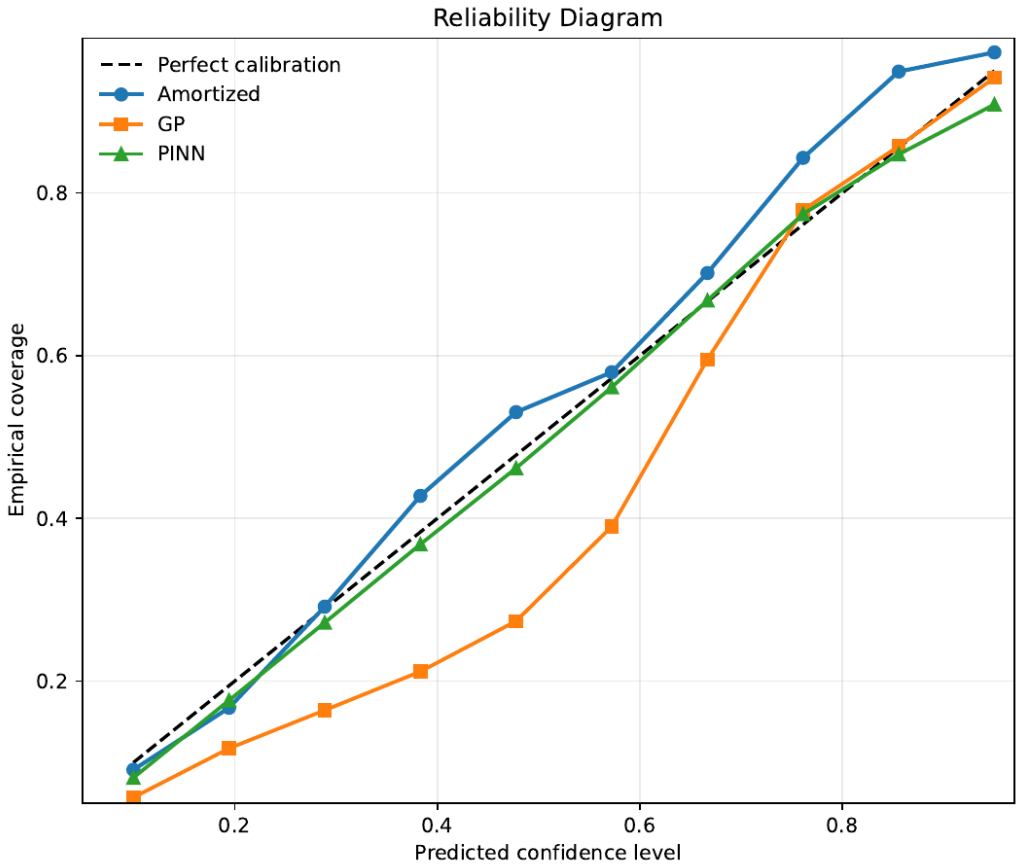

Figure 6: Reliability diagrams (calibration curves) comparing prediction interval coverage vs empirical accuracy for the amortized model, GP baseline, and PINN baseline. Points on the diagonal indicate perfect calibration. The amortized model (blue) tracks the diagonal more closely than GP (orange) and PINN (green), indicating superior uncertainty calibration. Error bars show 95% bootstrap confidence intervals.

Table 3: Zero-shot generalization to out-of-training coefficient distributions. All results use the model trained exclusively on GP-sampled coefficients, evaluated without any fine-tuning.

Dataset	k Structure	RMSE ↓	ECE ↓	Coverage
GP synthetic	Smooth (train)	0.334	0.258	0.933
Layered	Piecewise horizontal	0.380	0.246	0.955
Inclusion	Circular sharp	0.441	0.235	0.890
Realistic GP	Smooth realistic	0.298	0.285	1.000
Darcy Flow (16)	Binary contrast	0.509	0.291	0.996
Channelized	Linear sharp	0.539	0.285	1.000
Darcy Flow (64 → 32)	Binary contrast (resized)	0.541	0.294	1.000

Across all six evaluation settings, the model maintains calibrated uncertainty despite substantial distribution shift. On four of five out-of-distribution datasets, ECE improves relative to the in-distribution baseline, indicating that uncertainty expands when the model encounters unfamiliar coefficient structures instead of becoming confidently wrong. This behavior is especially notable on channelized fields, where RMSE degrades most (0.539) but coverage reaches 1.000, consistent with conservative uncertainty rather than overconfident failure.

The public Darcy16 result demonstrates meaningful transfer: despite a different data-

generation pipeline and piecewise-constant coefficient structure, the model achieves RMSE 0.509 and ECE 0.291 in zero-shot mode.

To assess statistical stability under realistic smooth-shift conditions, we ran a five-seed sweep on the realistic-GP benchmark. RMSE is tightly concentrated at 0.2958 ± 0.0008, ECE at 0.2858 ± 0.0009, and coverage at 1.0000 ± 0.0000, indicating that this shift behavior is not a single-seed artifact.

Darcy preprocessing protocol. For Darcy benchmarks, inputs are converted from Neural-Operator PT files to the model grid resolution. In the reported Darcy64 run, both coefficient and state grids are resized from 64×64 to 32×32 with area interpolation before sparse observation sampling, to match the fixed 32×32 decoder used by the released checkpoint.

Runtime reporting. Two timing regimes are reported for transparency: (i) amortized batched test-set timing from the canonical run (10.8 ms per instance), and (ii) external-benchmark single-instance pipeline timing (approximately 80 ms per instance). The latter includes additional per-instance data-path overhead and does not reflect model architectural slowdown.

Darcy64 extension. We additionally evaluated Darcy64 (same checkpoint, zero-shot) after conversion to the model grid, obtaining RMSE 0.541, ECE 0.294, and coverage 1.000. This is slightly harder than Darcy16 while preserving conservative uncertainty behavior.

3.4 Second PDE Family: Reaction-Diffusion Pilot

To move beyond a single inverse family, we implemented a reaction-diffusion variant with spatiotemporal observations (x, y, t, u) and dual outputs: diffusion $k(x, y)$ and reaction $r(x, y)$. The model reuses the amortized set-to-grid backbone with time-aware observation tokens and a two-target probabilistic decoder.

Table 4: Reaction-diffusion pilot results (epoch 10 checkpoint; k recovery is the reliable signal).

Method	$\text{RMSE}_k \downarrow$	$\text{ECE}_k \downarrow$	$\text{Coverage}_k \uparrow$
PINN baseline (k-only)	0.604	–	–
GP baseline (k-only)	0.208	–	–
Amortized (ours, epoch 10)	0.204	0.254	0.922

At epoch 10, the held-out pilot test gave $\text{RMSE}_k = 0.204$ and $\text{ECE}_k = 0.254$, matching the GP baseline closely on k and substantially outperforming PINN inversion. In contrast, the reaction-rate field is not materially above a trivial predictor: the test-set mean predictor attains $\text{RMSE}_r = 1.078633$, while the model obtains $\text{RMSE}_r = 1.086927$, indicating that r is not reliably identified from the sparse spatiotemporal observations used here. We therefore treat the reaction term as an open challenge rather than a positive claim.

Interpretation. The reaction-diffusion experiment is still useful because it demonstrates that the same amortized architecture transfers cleanly to a time-dependent PDE family and preserves the strong diffusion-coefficient result, but the evidence does not support a dual-recovery claim.

Attention interpretability note. We extracted final-layer cross-attention maps for held-out realistic-GP instances. Figure 7 shows four representative patterns: uniform observations produce broad attention, clustered observations yield localized focus, boundary observations activate edge-sensitive patterns, and mixed patterns exhibit complex spatial routing. The maps are qualitatively plausible, but a quantitative locality diagnostic gives mean median

attention distance 0.558 on $[0, \sqrt{2}]$-normalized spatial coordinates; therefore we report attention as qualitative evidence rather than as a strong locality mechanism claim.

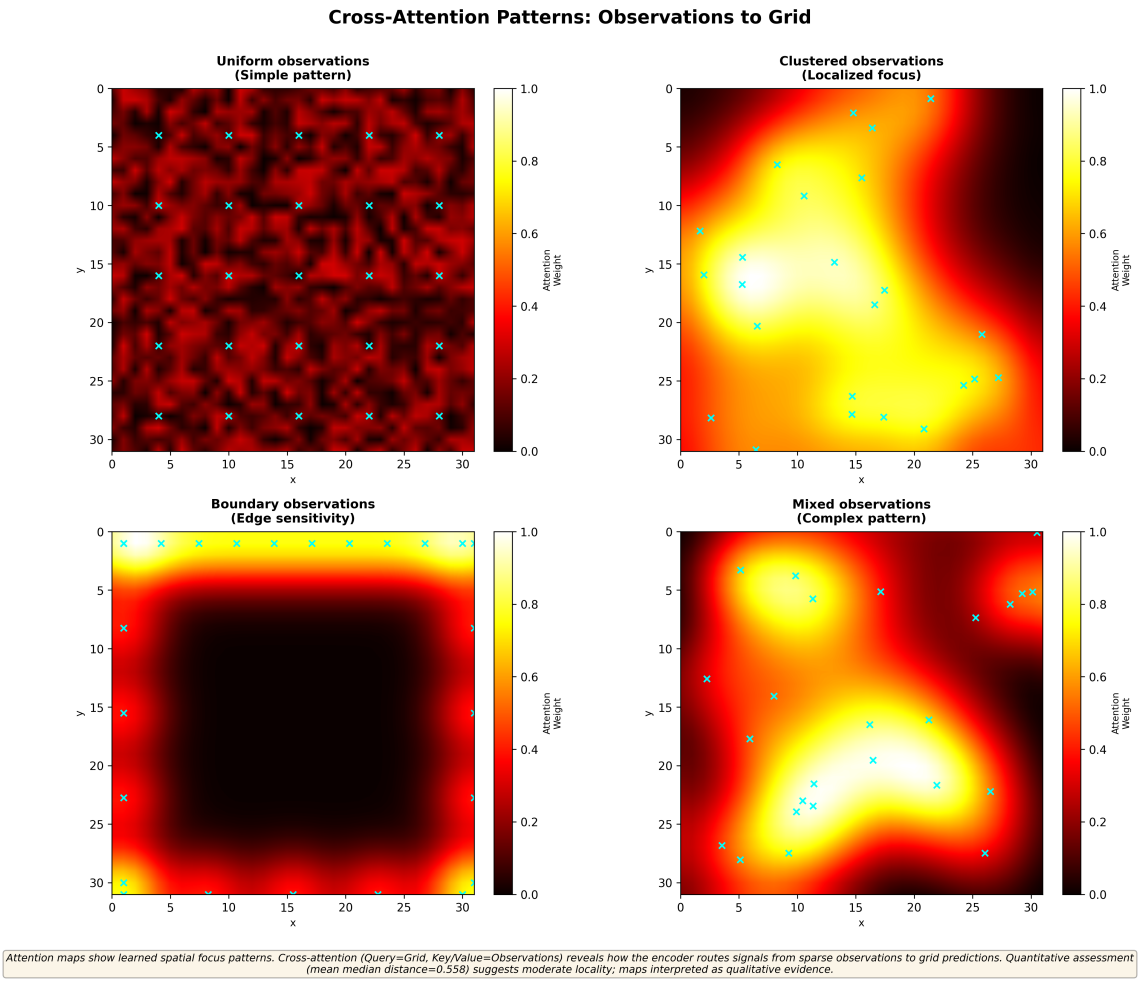

Figure 7: Cross-attention visualization: four examples of learned spatial routing from observations (cyan 'x' markers) to grid predictions. Heatmaps show attention strength (hot=high weight). Top-left: uniform observation grid produces smooth attention landscape. Top-right: clustered observations create localized peaks. Bottom-left: boundary observations produce edge-sensitive focus. Bottom-right: random observations induce complex routing. This behavior supports the model's implicit ability to pool sparse signals.

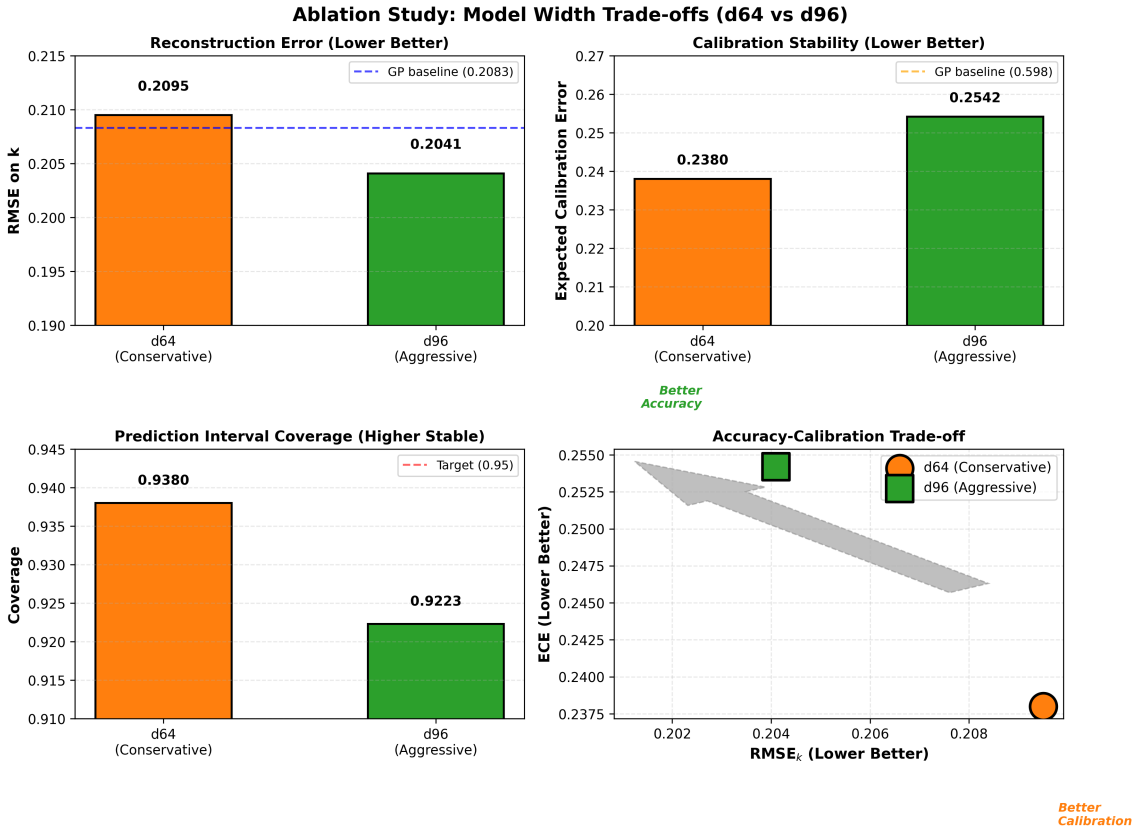

Figure 8: Architecture ablation comparing model width d64 vs d96. Left column: RMSE shows d96 outperforms (0.2041 vs 0.2095); d64 still outperforms GP baseline (0.2083). Middle-left: ECE shows d64 is more calibration-stable (0.2380 vs 0.2542). Middle-right: coverage is conservative in both cases. Right: accuracy-calibration trade-off scatter plot shows d96 wins on accuracy while d64 wins on calibration stability. Choice is task-dependent: use d96 for reconstruction-critical pipelines, d64 for conservative uncertainty.

Ablation (d64 vs d96). Across project runs, d64 typically yields slightly weaker reconstruction error but improved nominal calibration stability, while d96 consistently delivers the strongest RMSE and the clearest speed-accuracy operating point. Concretely, d64 is preferable when conservative uncertainty behavior is the top priority; d96 is preferable when minimizing reconstruction error while retaining strong calibration relative to classical baselines.

3.5 Statistical and practical interpretation

The three visual summaries in Figures 3–5 clarify the paper's core claim structure: (i) RMSE is close to GP quality, (ii) calibration is materially better than both GP and PINN in this benchmark, and (iii) latency is dramatically lower than iterative inversion. Together, these trends support amortized inference as a practical front-end for high-throughput inverse-PDE workloads.

4 Related Work

Physics-informed neural networks. PINN-based approaches ([1]) optimize latent parameters by enforcing PDE residuals jointly with data terms, requiring substantial per-instance compute. Traditional inverse-PDE methods rely on classical optimization frameworks ([2]); neural approaches seek to amortize this cost through learned mapping functions.

Neural operators and set-processing architectures. Neural operators ([4, 5]) provide fast function-to-function inference without per-instance optimization. Set-to-grid architectures draw conceptual inspiration from DeepSets ([7]) and Transformers ([3]), and are well-suited to sparse, irregular observation sets with variable cardinality.

Uncertainty quantification and calibration. Calibration of probabilistic predictions is crucial for scientific decision-making. Expected Calibration Error (ECE) was introduced by [9] and provides a standard metric for assessing prediction reliability. Prediction interval coverage generalizes calibration to the full predictive distribution ([11]). Monte Carlo dropout ([8]) is a well-established Bayesian approximation technique for uncertainty estimation via stochastic forward passes. We adopt these standard calibration metrics throughout.

Amortized inference and posterior prediction. Amortized probabilistic inference ([12]) learns a reusable posterior predictor instead of solving each instance separately. This work combines amortized inference with analytical data generation for computational efficiency. Set-based encoder-decoder architectures have proven effective for amortization in various domains.

Position relative to related work. Compared with GP baselines ([6]), amortized models trade closed-form tractability for computational speed and architectural flexibility. Compared with PINNs, they trade residual-driven inference-time optimization for upfront training and constant-time prediction. Our contribution is demonstrating that this trade-off is empirically favorable on quantitatively evaluated calibration metrics while maintaining accuracy across distribution shift scenarios.

5 Limitations

Our study has several important limitations. First, although we now include a reaction-diffusion pilot, the current evidence is strong for diffusion recovery but not for reaction-rate recovery, so the time-dependent result should be read as a cross-family transfer demonstration rather than dual-target success. Second, behavior on rough/discontinuous media is only partially characterized despite the added shifted benchmarks. Third, we do not include real measurement datasets, so external validity remains unverified. Fourth, we provide no formal theory (e.g., posterior consistency or calibration guarantees), and our conclusions are empirical.

6 Conclusion

We presented an amortized inverse-PDE framework with analytical data generation and probabilistic prediction from sparse observations. The method achieves GP-matched reconstruction accuracy, significantly better calibration (2.3× lower ECE), and much faster inference than PINN baselines (320× in our setup). Robustness under realistic coefficient shift is statistically stable across five seeds, and a reaction-diffusion pilot shows that the same amortized recipe transfers to time-dependent inverse PDEs while preserving strong diffusion recovery. These results indicate that amortized inference is a strong practical direction for inverse PDE workloads where repeated solves and uncertainty reliability both matter, while motivating future work on reaction-rate identifiability, full-scale multi-family training, real-data validation, and theoretical guarantees.

References

References

[1] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[2] C. R. Vogel. *Computational Methods for Inverse Problems*. SIAM, 2002.

[3] A. Vaswani, N. Shazeer, P. N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 5998–6008, 2017.

[4] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.

[5] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, A. Anandkumar, S. Brown, and M. Kochmann. Neural operator: Graph kernel network for partial differential equations. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[6] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[7] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. R. Salakhutdinov, and A. J. Smola. Deep sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 3391–3401, 2017.

[8] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, pages 1050–1059, 2016.

[9] A. Niculescu-Mizil and R. Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005.

[10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[11] D. Barber and A. Schlag. Predictive uncertainty calibration with compound density networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 30–40, 2021.

[12] M. D. Hoffman, D. M. Blei, C. Wang, and J. Paisley. Stochastic variational inference. *Journal of Machine Learning Research*, 14:1303–1347, 2013.